

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: df = pd.read_csv('supermarket_sales.csv')
```

```
In [3]: df.head()
```

```
Out[3]:
```

	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity	Tax 5%
0	750-67-8428	Alex	Yangon	Member	Female	Health and beauty	74.69	7	26.1415
1	226-31-3081	Giza	Naypyitaw	Normal	Female	Electronic accessories	15.28	5	3.8200
2	631-41-3108	Alex	Yangon	Normal	Female	Home and lifestyle	46.33	7	16.2155
3	123-19-1176	Alex	Yangon	Member	Female	Health and beauty	58.22	8	23.2880
4	373-73-7910	Alex	Yangon	Member	Female	Sports and travel	86.31	7	30.2085

```
In [4]: df.shape
df.dtypes
```

```
Out[4]: Invoice ID          object
Branch          object
City            object
Customer type   object
Gender          object
Product line    object
Unit price      float64
Quantity        int64
Tax 5%          float64
Sales           float64
Date            object
Time            object
Payment         object
cogs            float64
gross margin percentage float64
gross income    float64
Rating          float64
dtype: object
```

```
In [5]: df.isnull().sum()
```

```
Out[5]: Invoice ID          0
        Branch            0
        City              0
        Customer type     0
        Gender            0
        Product line      0
        Unit price        0
        Quantity          0
        Tax 5%            0
        Sales             0
        Date              0
        Time              0
        Payment           0
        cogs              0
        gross margin percentage 0
        gross income      0
        Rating            0
        dtype: int64
```

```
In [6]: df.duplicated().sum()
```

```
Out[6]: 0
```

```
In [7]: df['Date'] = pd.to_datetime(df['Date'])
        df['Time'] = pd.to_datetime(df['Time'])
```

C:\Users\dell\AppData\Local\Temp\ipykernel_14324\2029396998.py:2: UserWarning: Could not infer format, so each element will be parsed individually, falling back to `dateutil`. To ensure parsing is consistent and as-expected, please specify a format.

```
df['Time'] = pd.to_datetime(df['Time'])
```

```
In [9]: df.columns
```

```
Out[9]: Index(['Invoice ID', 'Branch', 'City', 'Customer type', 'Gender',
              'Product line', 'Unit price', 'Quantity', 'Tax 5%', 'Sales', 'Date',
              'Time', 'Payment', 'cogs', 'gross margin percentage', 'gross income',
              'Rating'],
              dtype='object')
```

```
In [10]: total_sales = np.sum(df['Sales'])
        avg_sales = np.mean(df['Sales'])

        total_sales, avg_sales
```

```
Out[10]: (322966.749, 322.966749)
```

```
In [12]: df.groupby('Product line')['Sales'].sum()
```

```
Out[12]: Product line
        Electronic accessories    54337.5315
        Fashion accessories       54305.8950
        Food and beverages        56144.8440
        Health and beauty         49193.7390
        Home and lifestyle        53861.9130
        Sports and travel         55122.8265
        Name: Sales, dtype: float64
```

```
In [13]: df.groupby('Gender')['Sales'].mean()
```

```
Out[13]: Gender
Female    340.931414
Male      299.055738
Name: Sales, dtype: float64
```

```
In [14]: df.groupby('Payment')['Sales'].sum()
```

```
Out[14]: Payment
Cash          112206.570
Credit card   100767.072
Ewallet        109993.107
Name: Sales, dtype: float64
```

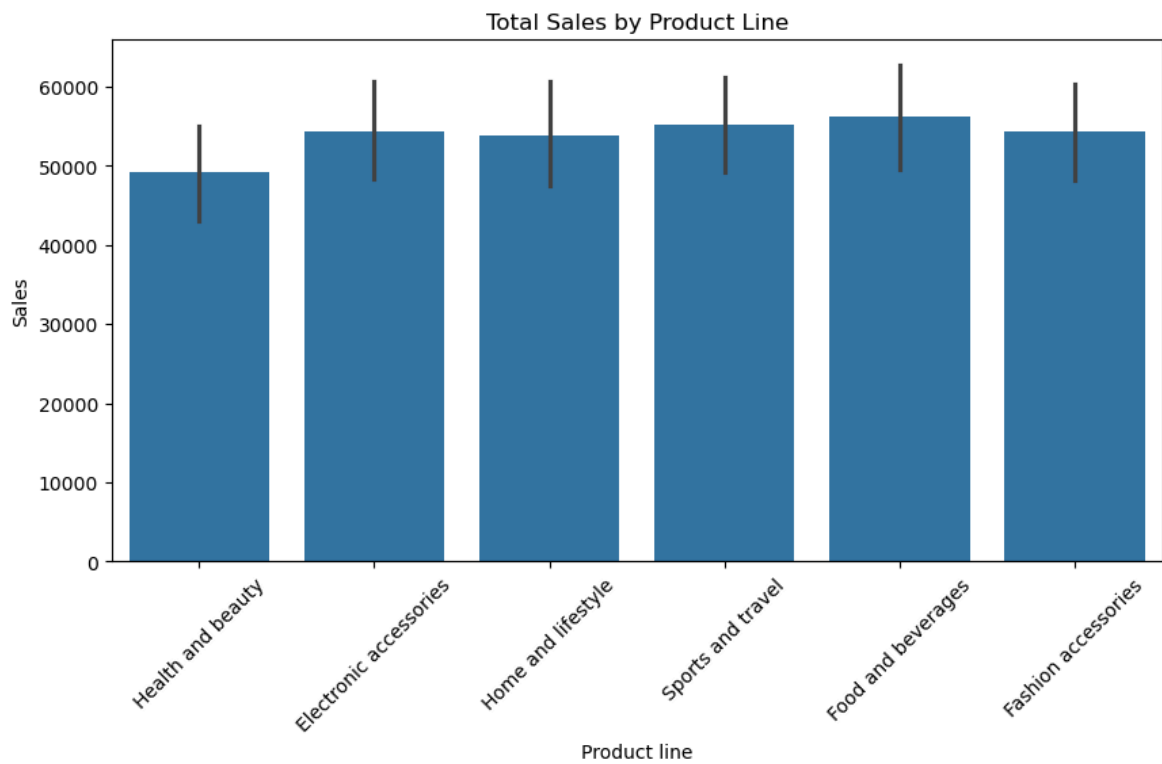
```
In [15]: df.isnull().sum()
```

```
Out[15]: Invoice ID          0
Branch          0
City            0
Customer type   0
Gender          0
Product line    0
Unit price      0
Quantity        0
Tax 5%          0
Sales           0
Date            0
Time            0
Payment         0
cogs            0
gross margin percentage  0
gross income    0
Rating          0
dtype: int64
```

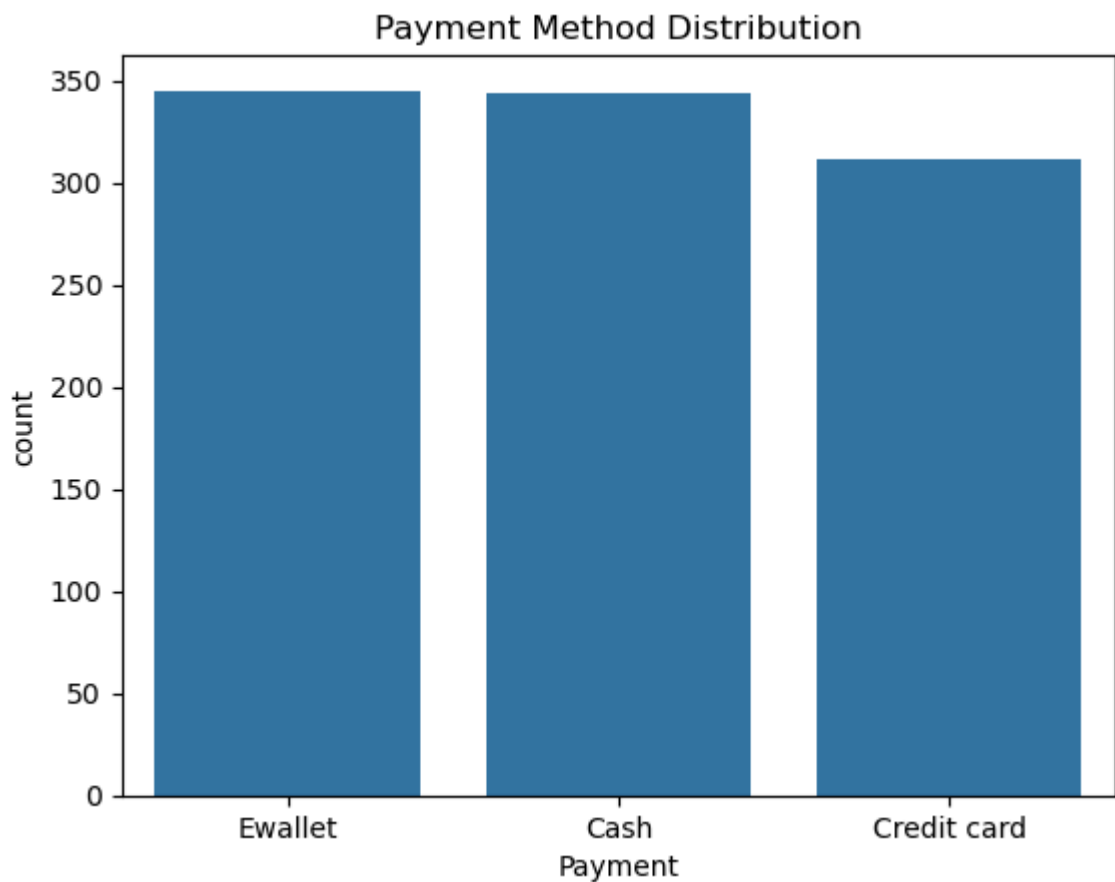
```
In [16]: df['Date'] = pd.to_datetime(df['Date'])
```

```
In [17]: df['Month'] = df['Date'].dt.month
df['Day'] = df['Date'].dt.day_name()
```

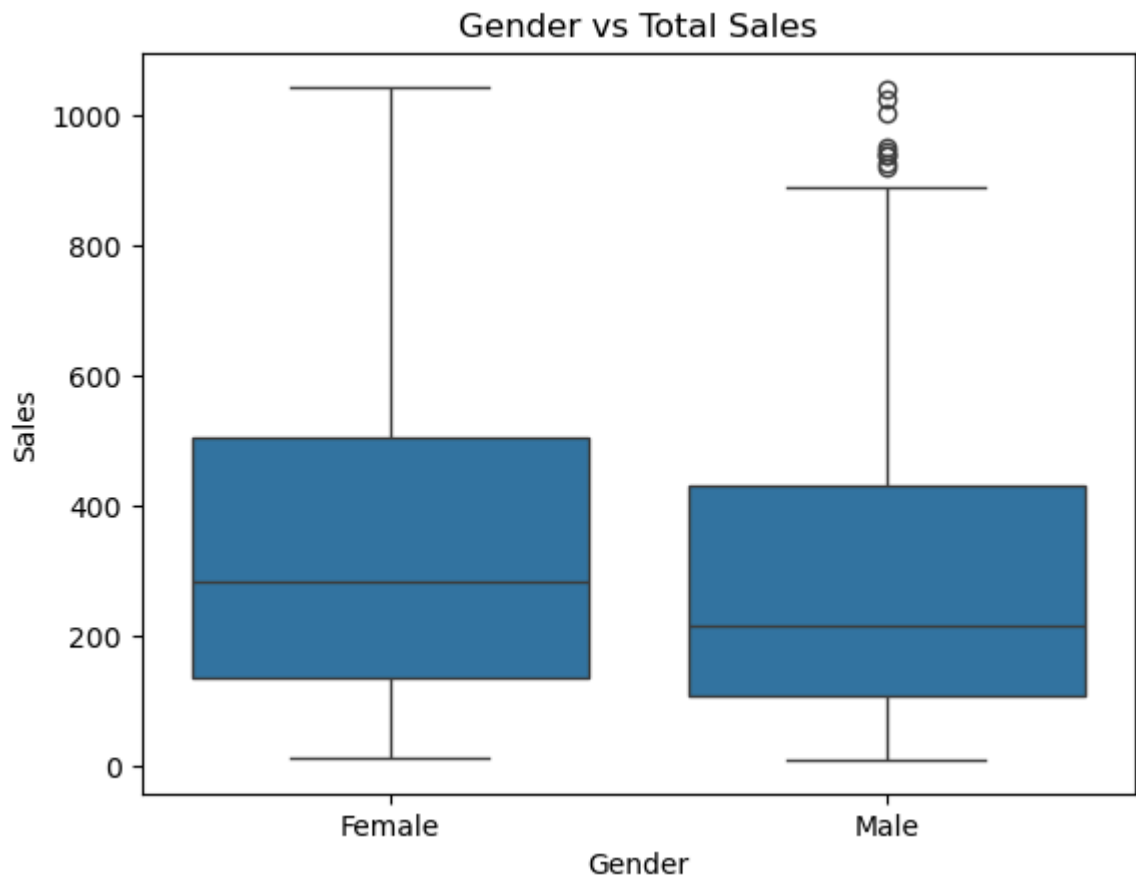
```
In [19]: plt.figure(figsize=(10,5))
sns.barplot(x='Product line', y='Sales', data=df, estimator=sum)
plt.xticks(rotation=45)
plt.title('Total Sales by Product Line')
plt.show()
```



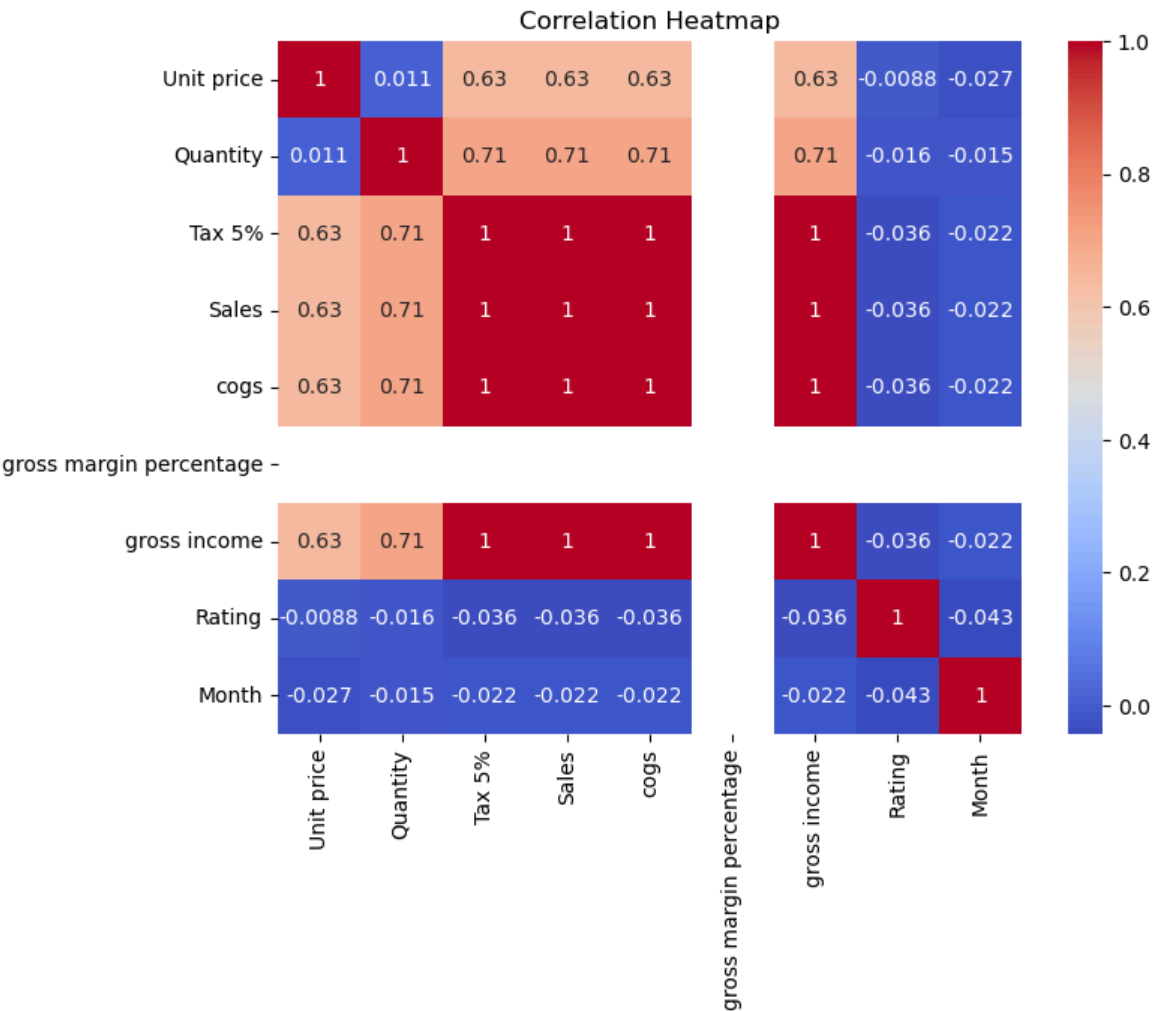
```
In [20]: sns.countplot(x='Payment', data=df)
plt.title('Payment Method Distribution')
plt.show()
```



```
In [22]: sns.boxplot(x='Gender', y='Sales', data=df)
plt.title('Gender vs Total Sales')
plt.show()
```



```
In [23]: plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```



```
In [ ]:
```